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				Date Prep. 26.03.2020		PROJECT TITLE				ABB EPMV		COVER SHEET		Scale		NTS		= H01			
				Prepared YR		BIODIESEL AND FLEXTech PLANT						BIODIESEL PLANT		Customer		PT Multimas Nabati Asahan		& 301			
00		Issued for Approval		26.03.20		YR		Checked AW		PT. ABB Sakti Industri				SCHEMATIC DIAGRAM		Drawing Nbr.		20.003-MNA-301		Sh. 1	
Ind.		Revision		Date		Name		Approved AI		Based on: BE320003				SBC-W INCOMING 1		Customer Doc. Nbr.				Cont. 29	
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UniSec

Medium Voltage Switchgear

PT MULTIMAS NABATI ASAHAN

BIODIESEL PLANT

SCHEMATIC DIAGRAM

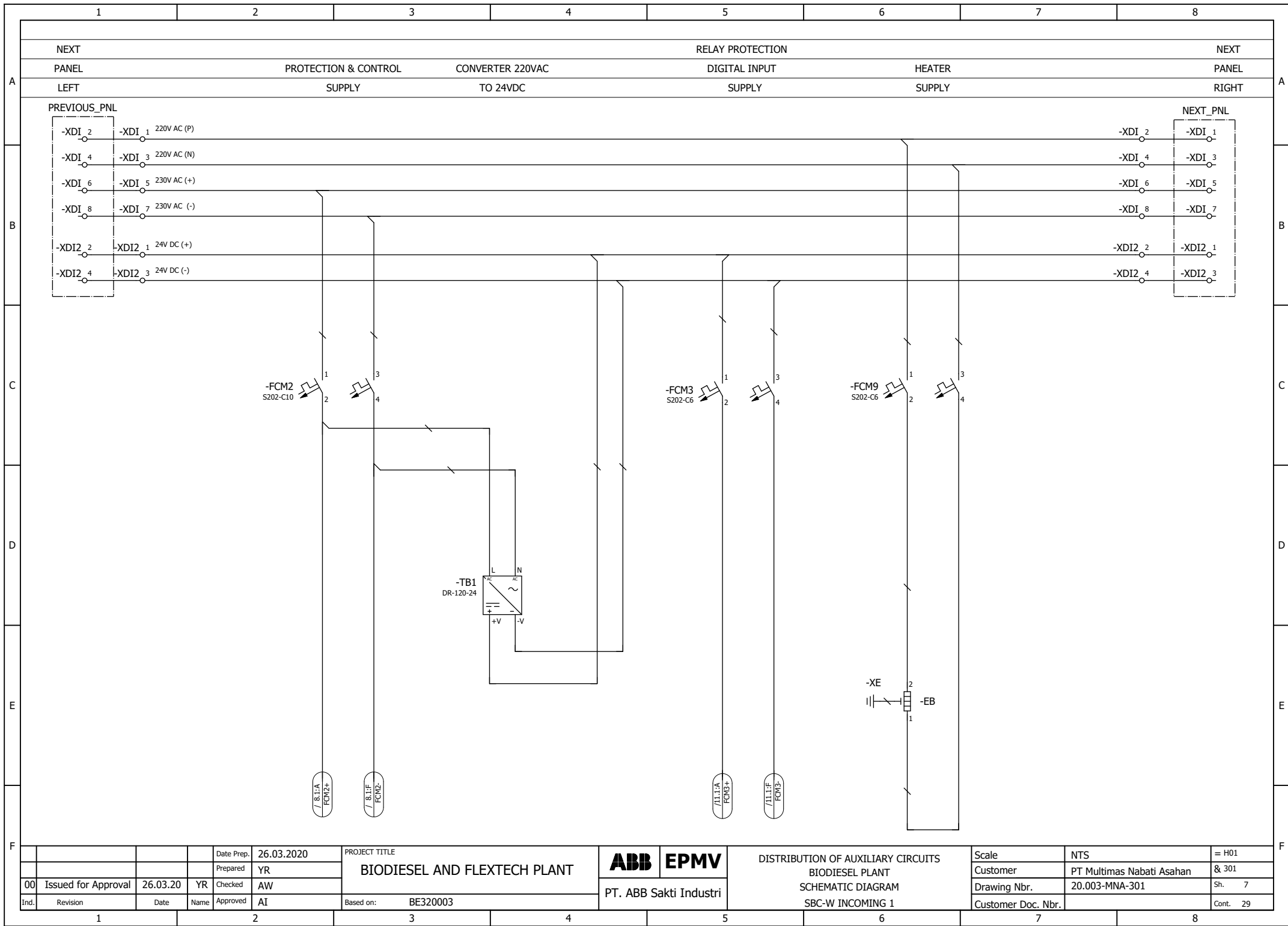
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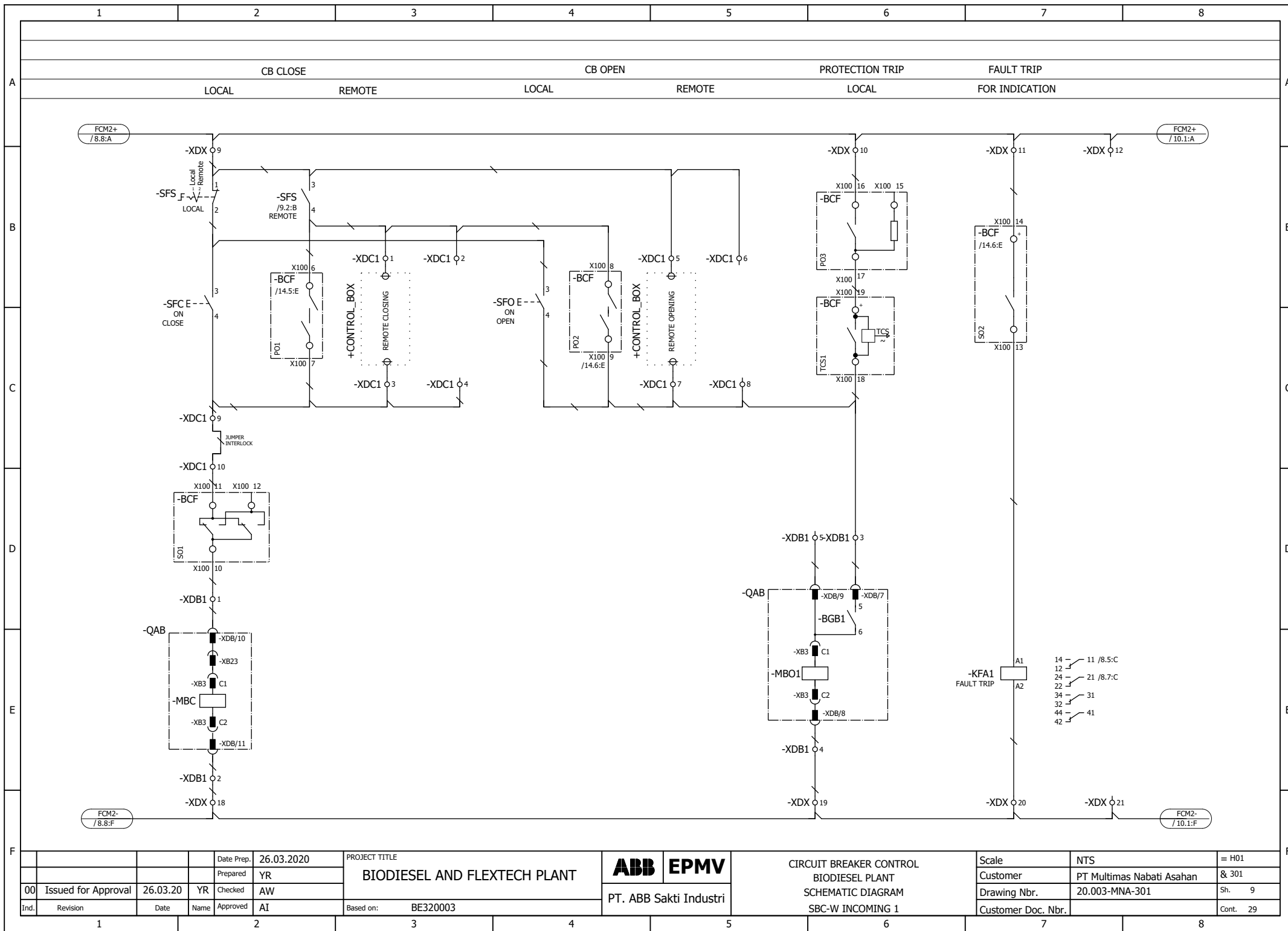
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A	-TFA	ACTIVE POWER TRANSDUCERS			-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS			A							
	-TFC	CURRENT TRANSDUCERS			-XD12	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING										
	-TFF	FREQUENCY TRANSDUCERS			-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION										
	-TFJ	ACTIVE ENERGY TRANSDUCERS			-XD14	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS										
	-TFK	REACTIVE ENERGY TRANSDUCERS			-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS										
	-TFM	MULTIFUNCTION TRANSDUCERS			-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS										
	-TFP	POWER-FACTOR TRANSDUCERS			-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS										
	-TFQ	REACTIVE POWER TRANSDUCERS			-XD19	TERMINAL BLOCK FOR SPECIAL USE										
	-TFS	SIGNAL CONVERTERS			-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD										
	-TFV	VOLTAGE TRANSDUCERS			-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS										
B	-WA	BUSBARS			-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT			B							
	-WBC	POWER CABLES			-XDS	CURRENT SOCKETS										
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.			-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB										
	-WGC	CONTROL CABLES			-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK										
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)			-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK										
	-WH	OPTICAL FIBRES			-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS										
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS			-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1										
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2										
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2			-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3										
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR										
C	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER			-XDX	SUPPORT TERMINAL BLOCK			C							
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS			-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE										
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)			-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE										
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE										
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS			-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE										
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS			-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE										
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE										
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT										
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR														
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET														
D	-XDC	CUSTOMER TERMINAL BLOCK							D							
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER														
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR														
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT														
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS														
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER														
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE														
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE														
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2														
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2														
E	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS							E							
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR														
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS														
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS														
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)														
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)														
	F				Date Prep.	26.03.2020	PROJECT TITLE			ABB	EPMV	REFERENCE DESIGNATIONS		Scale	NTS	= H01
				Prepared	YR	BIODIESEL AND FLEXTech PLANT		Customer	PT Multimas Nabati Asahan					& 301		
00		Issued for Approval	26.03.20	YR	Checked	AW	Drawing Nbr.	20.003-MNA-301	Sh. 6							
Ind.		Revision	Date	Name	Approved	AI	Based on:	BE320003		PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		SBC-W INCOMING 1		
												Customer Doc. Nbr.		Cont. 29		
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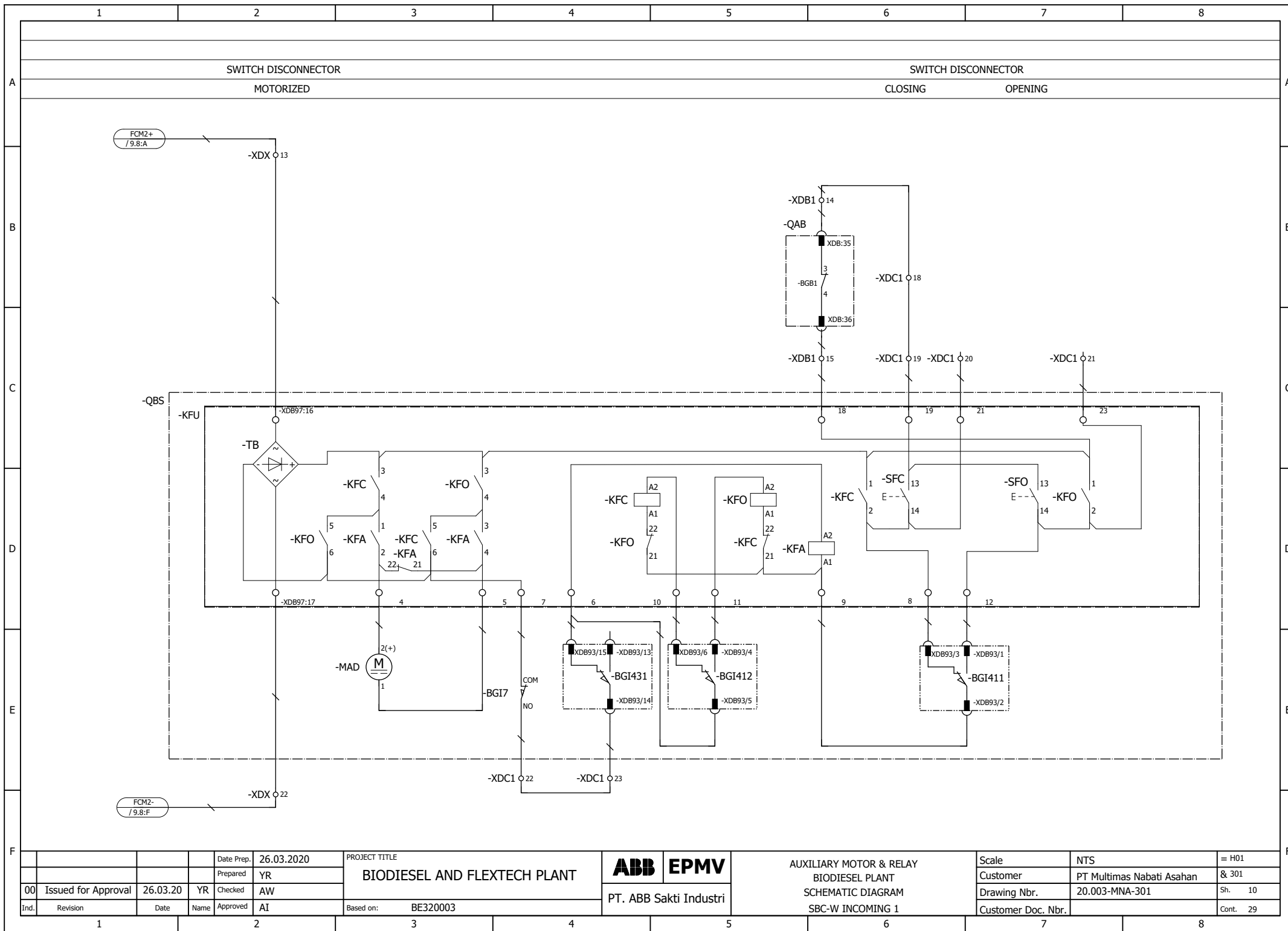
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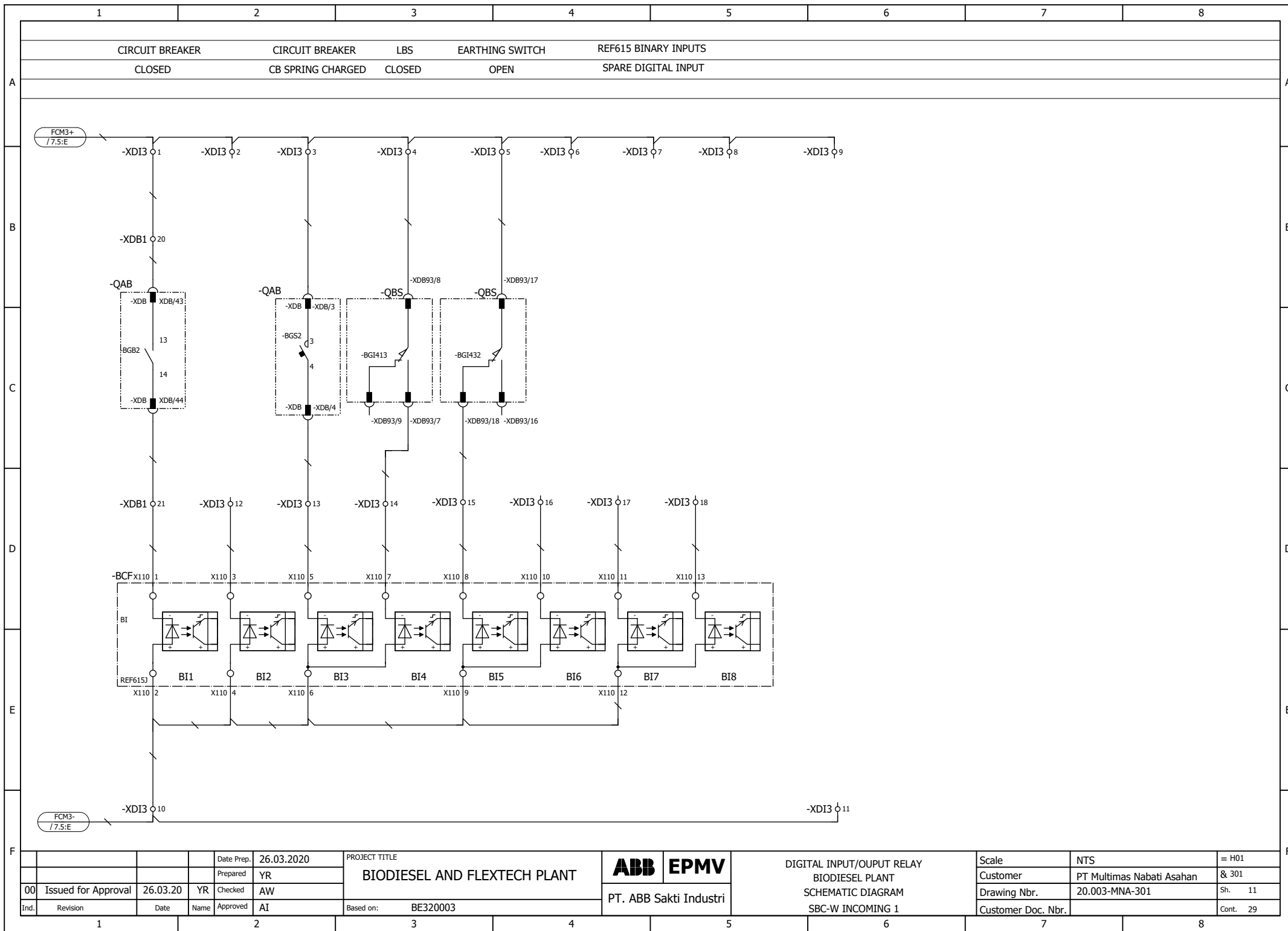
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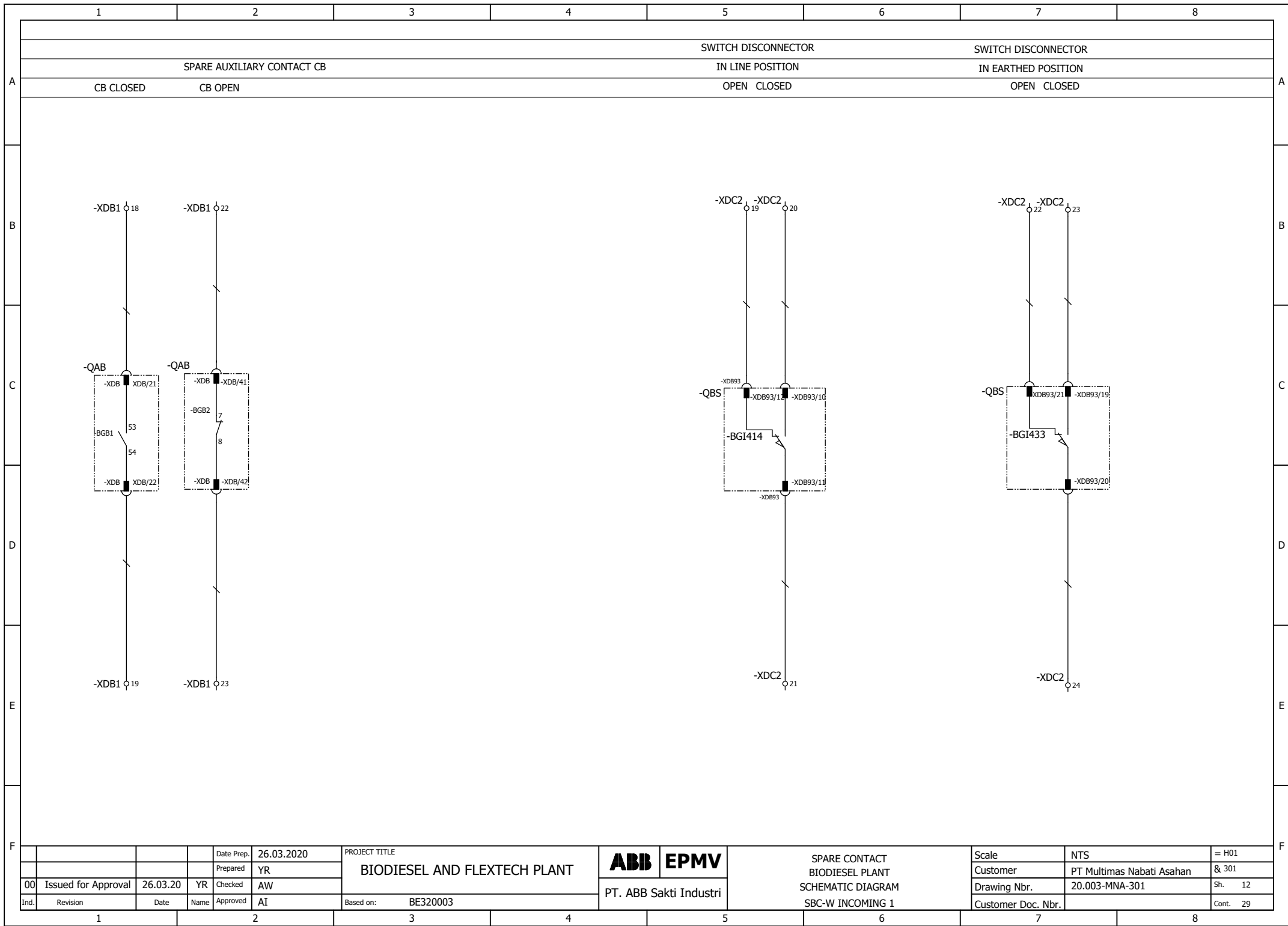
PROJECT TITLE	
BIODIESEL AND FLEXTech PLANT	
Based on: BE320003	

ABB	EPMV
PT. ABB Sakti Industri	

DIGITAL INPUT/OUTPUT RELAY
BIODIESEL PLANT
SCHEMATIC DIAGRAM
SBC-W INCOMING 1

Scale	NTS	= H01
Customer	PT Multimas Nabati Asahan	& 301
Drawing Nbr.	20.003-MNA-301	Sh. 11
Customer Doc. Nbr.		Cont. 29

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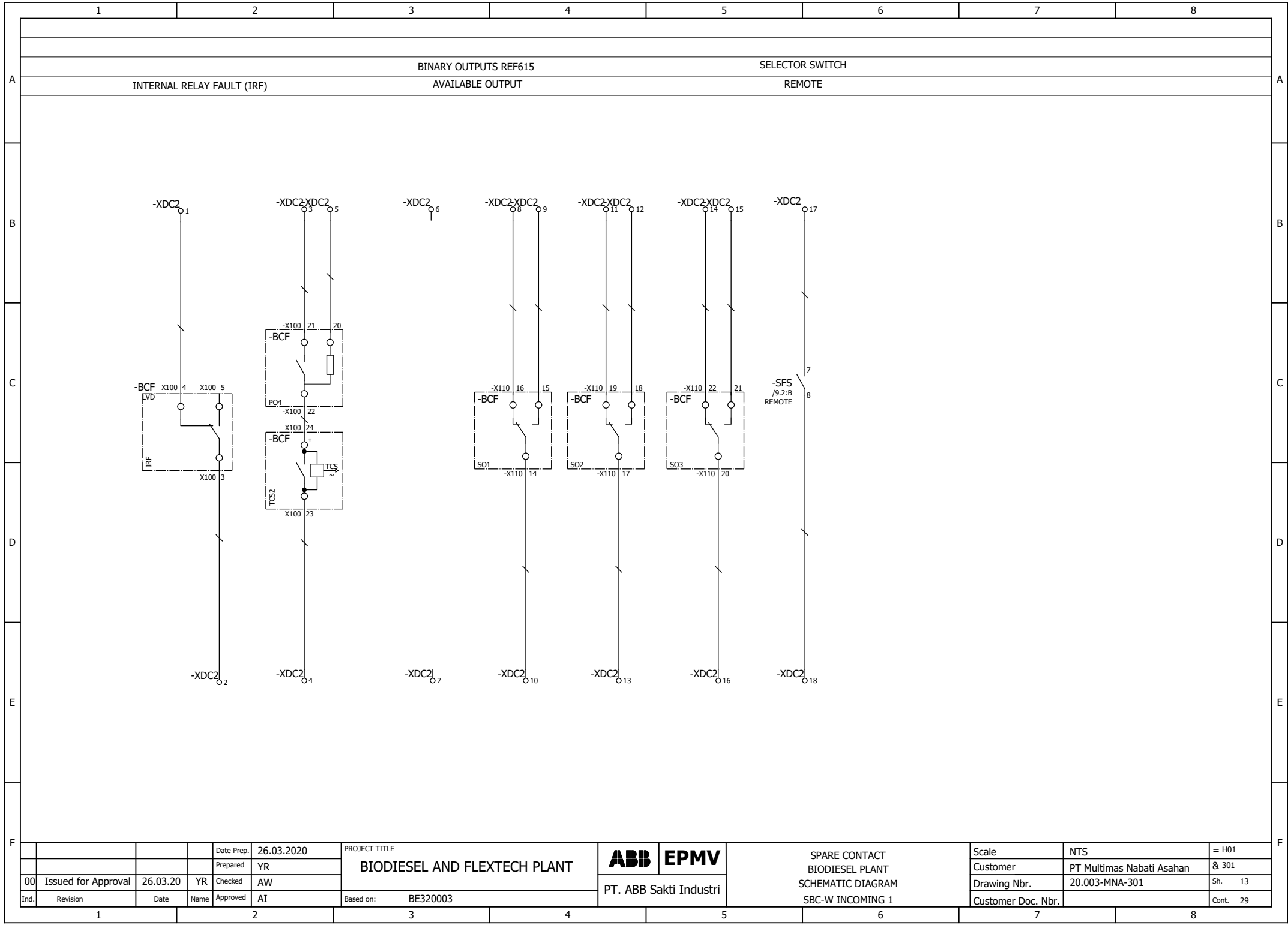
PROJECT TITLE	
BIODIESEL AND FLEXTech PLANT	
Based on:	BE320003

ABB	EPMV
PT. ABB Sakti Industri	

SPARE CONTACT
BIODIESEL PLANT
SCHEMATIC DIAGRAM
SBC-W INCOMING 1

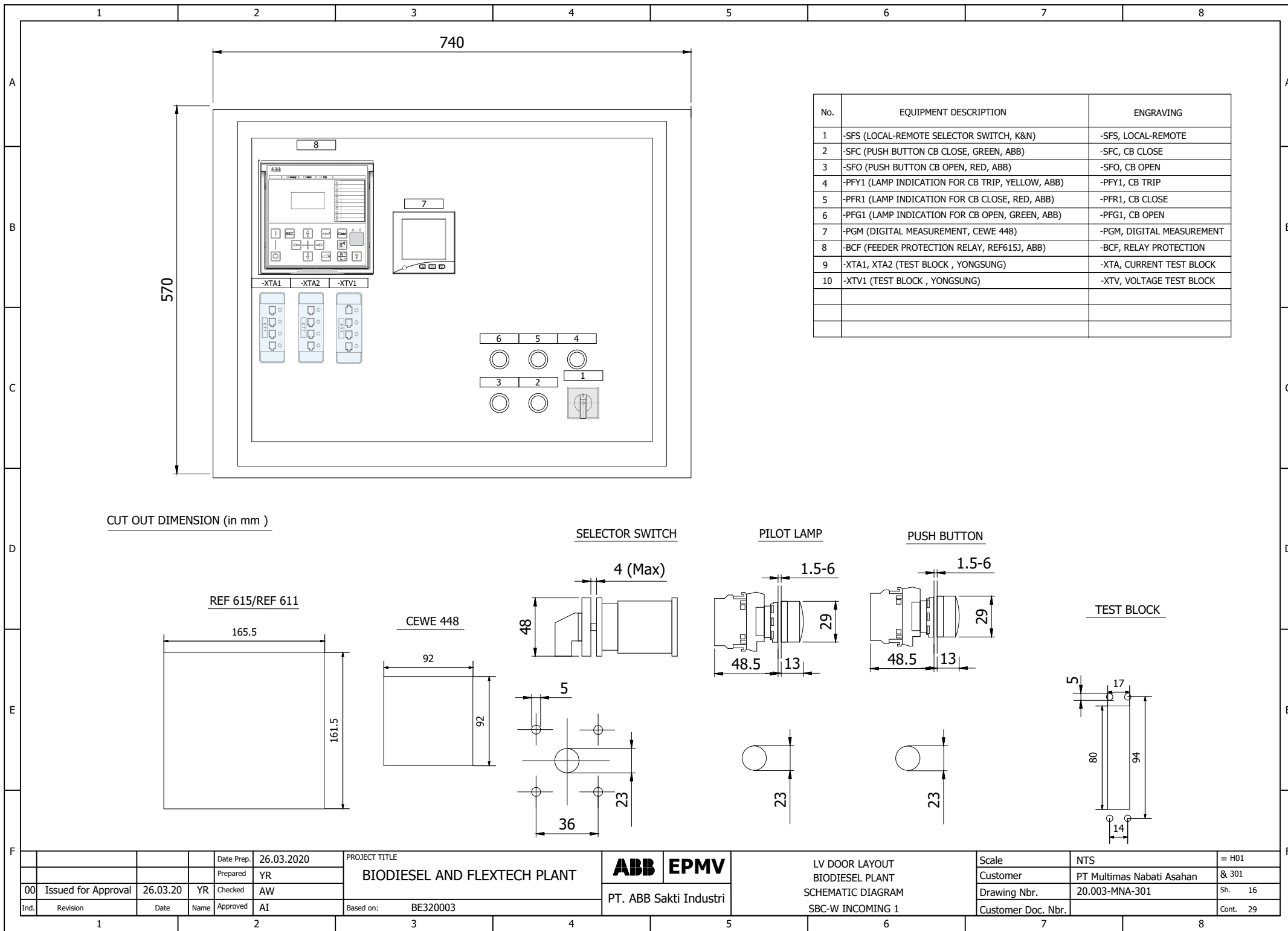
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Customer	PT Multimas Nabati Asahan	& 301
Drawing Nbr.	20.003-MNA-301	Sh. 12
Customer Doc. Nbr.		Cont. 29

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A	BINARY OUTPUTS REF615				SELECTOR SWITCH																																												
	INTERNAL RELAY FAULT (IRF)		AVAILABLE OUTPUT		REMOTE																																												
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F	<table border="1"> <tr> <td></td><td></td><td>Date Prep.</td><td>26.03.2020</td><td colspan="2">PROJECT TITLE</td><td rowspan="2">ABB EPMV</td><td rowspan="2">SPARE CONTACT BIODIESEL PLANT SCHEMATIC DIAGRAM SBC-W INCOMING 1</td><td>Scale</td><td>NTS</td><td>= H01</td></tr> <tr> <td></td><td></td><td>Prepared</td><td>YR</td><td colspan="2">BIODIESEL AND FLEXTECH PLANT</td><td>Customer</td><td>PT Multimas Nabati Asahan & 301</td></tr> <tr> <td>00</td><td>Issued for Approval</td><td>26.03.20</td><td>YR</td><td>Checked</td><td>AW</td><td rowspan="2">PT. ABB Sakti Industri</td><td rowspan="2"></td><td>Drawing Nbr.</td><td>20.003-MNA-301</td><td>Sh. 13</td></tr> <tr> <td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE320003</td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 29</td></tr> </table>										Date Prep.	26.03.2020	PROJECT TITLE		ABB EPMV	SPARE CONTACT BIODIESEL PLANT SCHEMATIC DIAGRAM SBC-W INCOMING 1	Scale	NTS	= H01			Prepared	YR	BIODIESEL AND FLEXTECH PLANT		Customer	PT Multimas Nabati Asahan & 301	00	Issued for Approval	26.03.20	YR	Checked	AW	PT. ABB Sakti Industri		Drawing Nbr.	20.003-MNA-301	Sh. 13	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE320003	Customer Doc. Nbr.		Cont. 29
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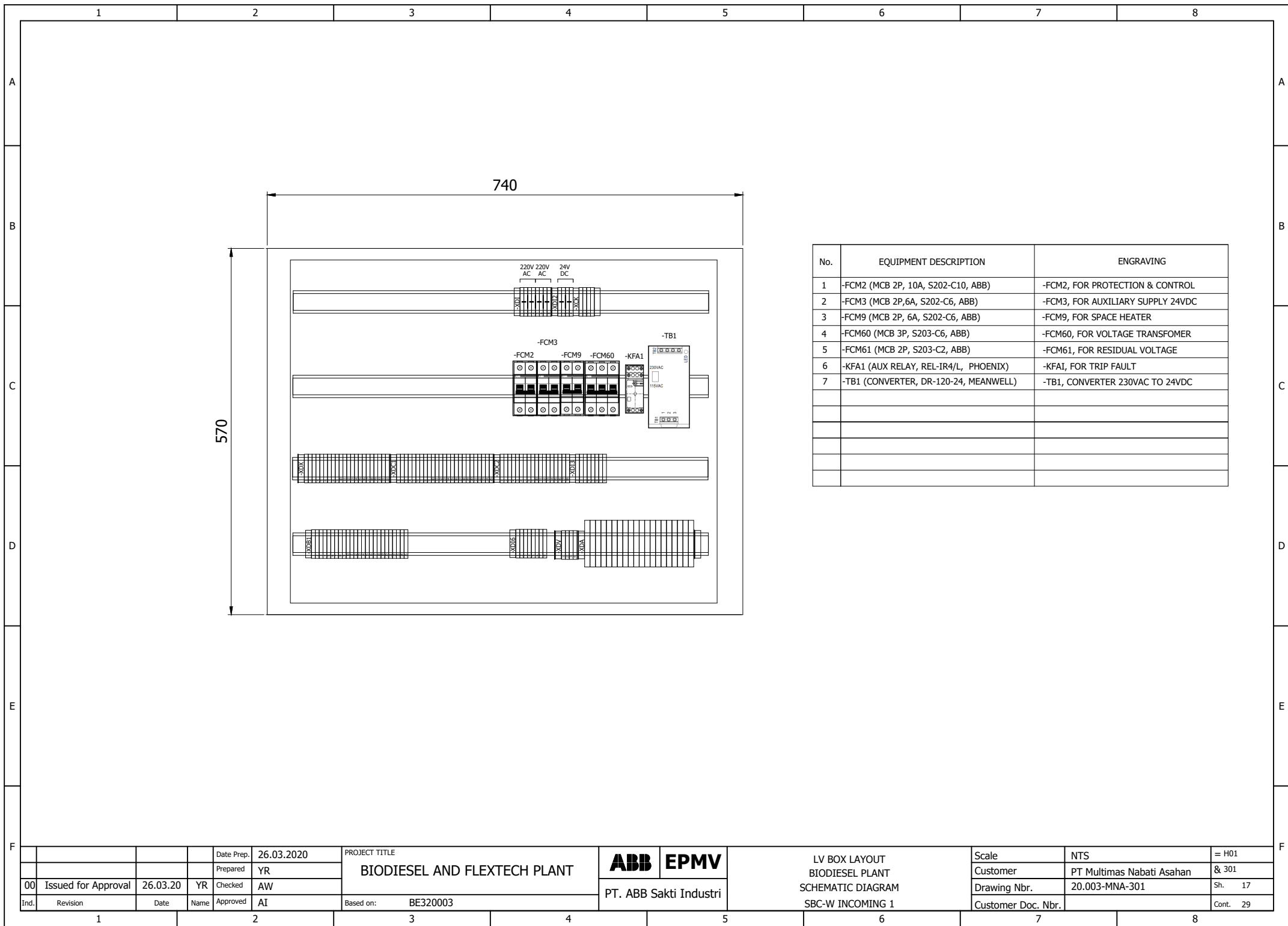
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			Prepared	YR					Customer	PT Multimas Nabati Asahan	& 301	
00	Issued for Approval	26.03.20	YR	Checked		AW	PT. ABB Sakti Industri		Drawing Nbr.	20.003-MNA-301	Sh.	14
Ind.	Revision	Date	Name	Approved	AI	Based on: BE320003			Customer Doc. Nbr.		Cont.	29

			Date Prep.	26.03.2020	PROJECT TITLE BIO DIESEL AND FLEXTech PLANT	 	VOLTAGE TRANSFORMER BIO DIESEL PLANT SCHEMATIC DIAGRAM SBC-W INCOMING 1	Scale	NTS	= H01
			Prepared	YR				Customer	PT Multimas Nabati Asahan	& 301
00	Issued for Approval	26.03.20	YR	Checked				AW	Drawing Nbr.	20.003-MNA-301
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Summarized parts list															IDABB_F02_005-IDABB				
															Carry over: 0.00				
A	TAG ID				DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS				
	-BCF				Feeder Management Relay HBFJAEAGABC1ANA11G				ABB		REF615J HBFJAEAGABC1ANA11G		2		/8.2:C;/15.7:B				
B	-EB				HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.6:E				
	-FCM2				MCB S202-C10 MCB. 2P, 10A				ABB		2CDS252001R0104		1		/7.2:C				
	-FCM3;-FCM9				MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		2		/7.5:C;/7.6:C				
	-FCM60				MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/15.4:A				
	-FCM61;-FCM62				MCB S202-C2 MCB. 2P, 2A				ABB		2CDS252001R0024		2		/15.5:D;/15.7:E				
C	-KFA1				Auxiliary Relay 230AC 4COS				PHOENIX		1YRA3DL266230001		1		/9.7:E				
	-PFG1				PILOT LIGHT GREEN #CL-523G#230VAC CL-523G				ABB		1SFA619402R5232		1		/8.5:E				
	-PFR1				PILOT LIGHT RED #CL-523R#230VAC CL-523R				ABB		1SFA619402R5231		1		/8.4:E				
	-PFY1				PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y				ABB		1SFA619402R5233		1		/8.6:E				
	-PGM1				POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC				CEWE		1YRA4EA0350001		1		/8.3:C				
D	-SFC				PUSH BUTTON, CP1-10G-11,1NC+1NO GREEN COLOUR				ABB		1SFA619100R1072		1		/9.2:B				
	-SFO				PUSH BUTTON, CP1-10R-11,1NC+1NO RED COLOUR				ABB		1SFA619100R1071		1		/9.4:B				
	-SFS				LOCAL/REMOTE Selector Switch Type : S0-CA10-A721-600-FT2; Contact : 2NC & 2NO; Mounting Type : FT2				K&N		1YRA3DA17900201		1		/9.2:B				
	-TB1				Voltage Input : 85-264 VAC/120-370V DC Voltage Output : 24 VDC, 5A				MEANWELL		DR-120-24		1		/7.3:D				
	-XTA1;-XTA2				Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSCTT-04C		2		/14.4:B;/14.4:D				
E	-XTV1				Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSPTT-04C		1		/15.5:A				
F																			
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XCK

REMARK: ☆ INTERPANEL LIGHT BUS OFFERED TOGETHER WITH PANELS.
 ▲ OUTGOING CONTROL CABLE (Supply by Customer).
 ● BRIDGE BAR.
 ✕ PARTITION PLATE.
 (TEST JUMPER.

REMARK: ☆ INTERPANEL LIGHT BUS OFFERED TOGETHER WITH PANELS.

- △ OUTGOING CONTROL CABLE (Supply by Customer).
- BRIDGE BAR.
- ✕ PARTITION PLATE.
- (TEST JUMPER.

REMARK: ☆ INTERPANEL LIGHT BUS OFFERED TOGETHER WITH PANELS.

- △ OUTGOING CONTROL CABLE (Supply by Customer).
- BRIDGE BAR.
- ✕ PARTITION PLATE.
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